

Chapter 37: Puerperal Infection — Revision Notes

Williams Obstetrics 26e, pp. 1673–1713 of the PDF. High-yield numbers are in **bold**. The five tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 8 figures are included from the PDF (Figure 37-7 appears twice in the PDF, on pp. 1696 and 1697; it is included once).

Big picture: Route of delivery is the single biggest risk factor: cesarean carries far more infection than vaginal birth. Metritis is **polymicrobial** and **~90% respond within 48–72 h**. Persistent fever means a search for **wound infection, phlegmon, abscess/hematoma or septic pelvic thrombophlebitis**, not resistant bacteria. **Single-dose prophylaxis at cesarean** has done more than anything else to cut postcesarean infection.

1. Puerperal Pelvic Infections — Overview

- **Puerperal infection** = any bacterial infection of the genital tract after delivery.
- Historically, the "lethal triad" of maternal death was **infection, preeclampsia and hemorrhage**.
- US Pregnancy Mortality Surveillance, **2006–2010 (4693 deaths)**: infection caused **13.6%** of deaths and was the **2nd leading cause**. In North Carolina, **40%** of infection-related maternal deaths were **preventable**.
- Postpartum hospitalization for **sepsis is rising**.

Puerperal fever

- **Definition: $\geq 38.0^{\circ}\text{C}$ (100.4°F).**
- Fever in the first 24 h → pelvic infection in only **~20% after vaginal birth**, but **~70% after cesarean**.
- **Most persistent postpartum fevers are genital tract infections.**
- **⚠ Spiking fever $\geq 39^{\circ}\text{C}$ in the first 24 h** → think **group A streptococcus** (virulent pelvic infection).

Other cause of fever	Key points
Breast engorgement	~15% of women who do not breastfeed . Rarely $>39^{\circ}\text{C}$, first few days, lasts <24 h . Less common in breastfeeding women.
Urinary infection	Uncommon (normal postpartum diuresis). Pyelonephritis: fever first, then CVA tenderness, nausea, vomiting.
Wounds	Episiotomy, perineal lacerations, abdominal incisions
Atelectasis (after general anesthesia)	Hypoventilation. Prevent with scheduled coughing and deep breathing . Fever comes from infection of normal flora distal to obstructing mucus plugs.

2. Uterine Infection (Metritis)

- Also called endometritis, endomyometritis, endoparametritis. Williams prefers **"metritis with pelvic cellulitis."**
- After vaginal birth: infection of the **decidua, myometrium and parametrium**. After cesarean: essentially a **surgical-site infection of the incised myometrium**.

Predisposing factors

Route of delivery — the single most significant risk factor

Setting	Metritis rate
Vaginal delivery (Parkland)	1–2%
Vaginal + ROM, prolonged labor, multiple cervical exams	5–6%
Intrapartum chorioamnionitis	>13% persistent uterine infection
MFMU cohort (>115,000) overall pelvic infection	~5%
Cesarean, 1970s Parkland, no prophylaxis	50%

- Cesarean: higher rehospitalization for wound complications and metritis. French enquiry: infection-related mortality **~25-fold** higher with cesarean.
- **Operative vaginal delivery:** ↑ metritis and perineal infection vs spontaneous birth.
 - Knight 2019 (**3500** women): **single-dose prophylactic antibiotic** ↓ maternal infection (urinary, uterine, perineal).
 - But **~50%** had "failure to progress" and **10%** had ROM 24–48 h — these confounders need study before routine adoption.
- **Cesarean prophylaxis:** a **single perioperative dose for all cesareans** (ACOG) → pelvic infection ↓ **65–75%**. **β-lactams are superior.**

Other risk factors

- **Postcesarean:** prolonged labor, membrane rupture, multiple cervical exams, **internal fetal monitoring.**
- **Any route:** lower socioeconomic status. Anemia and poor nutrition rarely matter (except extreme cases).
- **Lower genital tract colonization:** **GBS, Chlamydia trachomatis, Mycoplasma hominis, Ureaplasma urealyticum, Gardnerella vaginalis.**
- Also: general anesthesia, large hysterotomy extension, young age, nulliparity, prolonged induction, chorioamnionitis, obesity, meconium-stained fluid.

Microbiology

- **Most infections are polymicrobial** (bacterial synergy). Hematomas and devitalized tissue promote virulence.
- The uterine cavity is **usually sterile until the amnionic sac ruptures**; labor and manipulation then contaminate fluid and uterus.

Table 37-1. Bacteria Commonly Responsible for Female Genital Infections

Group	Organisms
Aerobes	
Gram-positive cocci	Group A, B, and D streptococci, enterococcus, <i>Staphylococcus aureus</i> , <i>Staphylococcus epidermidis</i>
Gram-negative bacteria	<i>Escherichia coli</i> , <i>Klebsiella</i> , <i>Proteus</i> spp.
Gram-variable	<i>Gardnerella vaginalis</i>
Others	<i>Mycoplasma</i> spp., <i>Chlamydia trachomatis</i> , <i>Neisseria gonorrhoeae</i>
Anaerobes	
Cocci	<i>Peptostreptococcus</i> , <i>Peptococcus</i> spp.
Others	<i>Clostridium</i> , <i>Bacteroides</i> , <i>Fusobacterium</i> , <i>Mobiluncus</i> spp.

Amnionic fluid cultures at cesarean (labor, ROM >6 h, before prophylaxis; Gilstrap 1979)

- **All** grew bacteria; average **2.5 organisms** per specimen.
- **Mixed aerobic + anaerobic 63%**, anaerobes alone **30%**, aerobes alone **7%**.
- Anaerobes: **Peptostreptococcus/Peptococcus 45%**, *Bacteroides* 9%, *Clostridium* 3% (rare but can be severe).
- Aerobes: **Enterococcus 14%**, *E. coli* 9%, GBS 8%.
- Isolates at cesarean match those from metritis on day 3. Commonest **blood culture** isolates: **GBS, E. coli, enterococci.**

Specific organisms

Organism	Point
MRSA	Now prevalent in skin/soft-tissue infection; more often abdominal and perineal wound infection than metritis
Group A β-hemolytic streptococcus	1 in 1220 births. Toxic shock–like syndrome. Onset before, during or within 12 h of delivery → maternal mortality ~90% , fetal mortality >50% (older reviews)
<i>Ureaplasma urealyticum</i>	Heavy cervical colonization may contribute. Azithromycin-based extended prophylaxis ↓ postcesarean infection 12% → 6% vs β -lactam alone
<i>Chlamydia</i>	Late-onset, indolent metritis
Bacterial vaginosis (early pregnancy)	3-fold ↑ puerperal infection (Sweden)

Cultures

- **Routine genital tract cultures before treatment are of little use** and add cost.
- Temperature **>102°F** → bacteremia more likely, but **routine blood cultures seldom change care.**
 - Before prophylaxis era: positive in **13%** (Parkland) and **24%** (LA County) with postcesarean metritis. Finland later: **5%** with puerperal sepsis.
 - Exception: very high spikes suggesting **group A strep.**

Pathogenesis and clinical course

- **Vaginal birth:** placental implantation site, decidua and adjacent myometrium, or cervicovaginal lacerations.
- **Cesarean:** an infected surgical incision. Cervicovaginal bacteria enter the uterus in labor and invade devitalized tissue after delivery.
- **Parametrial cellulitis** = infection of the retroperitoneal fibroareolar pelvic tissue. Usually contained if treated early; may spread deep.
- **Fever is the most important diagnostic criterion.**
 - Usually **38–39°C**; **>39°C suggests bacteremia or endotoxemia.**
- Abdominal pain; **parametrial tenderness** on abdominal and bimanual exam.
- **WBC 15,000–30,000/ μ L** (cesarean itself raises WBC).
- **Foul lochia is unreliable** (can occur without infection and vice versa). ⚠ **Group A strep may give scant, odorless lochia.**

Treatment

- **Mild metritis after vaginal birth:** oral or IM antibiotic may suffice.
- **Moderate–severe: IV broad-spectrum** regimen.
- **~90% improve within 48–72 h.**
- **Persistent fever beyond this** → search for:
 - **Parametrial phlegmon** (intense cellulitis)
 - **Abdominal incisional or pelvic abscess, or infected hematoma**
 - **Septic pelvic thrombophlebitis**
 - ⚠ Rarely resistant bacteria or drug fever.
- **Discharge after ≥ 24 h afebrile; no oral antibiotics needed afterward.**

Choice of antimicrobials

- Empirical, against mixed flora.
- **After vaginal delivery:** up to **90%** respond to **ampicillin + gentamicin.**
- **After cesarean:** include **anaerobic coverage.**

Table 37-2. Antimicrobial Regimens for Pelvic Infections Following Cesarean Delivery

Regimen	Comments
Clindamycin + gentamicin	"Gold standard," 90–97% efficacy , once-daily gentamicin dosing acceptable
Plus	Ampicillin added to regimen with sepsis or suspected enterococcal infection
Clindamycin + aztreonam	Gentamicin substitute for renal insufficiency
Extended-spectrum penicillins	Piperacillin, piperacillin/tazobactam, ampicillin/sulbactam, ticarcillin/clavulanate
Cephalosporins	Cefotetan, cefoxitin, cefotaxime, ceftriaxone
Vancomycin	Added to other regimens for suspected <i>Staphylococcus aureus</i> infections
Metronidazole + ampicillin + gentamicin	Metronidazole has excellent anaerobic coverage
Carbapenems	Imipenem/cilastatin, meropenem, ertapenem; all reserved for special indications

- **DiZerega 1979:** clindamycin + gentamicin **95%** response vs penicillin G + gentamicin — still the standard.
- **Enterococcus** may persist → add **ampicillin** initially or after **48–72 h** without response.
- **Gentamicin levels:** Parkland does **not** monitor routinely if **creatinine is normal**. **Once-daily = multiple-daily dosing** in levels and cure.
- **Reduced GFR:** clindamycin + 2nd-generation cephalosporin, or **clindamycin + aztreonam** (monobactam, aminoglycoside-like activity) — avoids gentamicin nephro-/ototoxicity.
- **β-lactams:** safe except allergy; β-lactamase inhibitors (**clavulanic acid, sulbactam, tazobactam**) extend spectrum.
- **Metronidazole:** superior anaerobic activity; second line for *C. difficile* colitis.
- **Imipenem + cilastatin** (cilastatin blocks renal metabolism): broad but costly → reserve for serious **nonobstetrical** infections.
- **Vancomycin:** gram-positive only; use for **type 1 (anaphylactic) β-lactam allergy**, suspected *S. aureus*, and *C. difficile*.

Perioperative prophylaxis

Table 37-3. Various Prophylactic Methods for Decreasing Pelvic and Wound Infection Rates Following Delivery

Route	Method	Study Results
Routine delivery	Peripartum antimicrobials	Limited evidence, may reduce risk (Bonet, 2017a)
Episiotomy	Perioperative prophylaxis	Insufficient evidence (Bonet, 2017b)
Operative vaginal delivery	Peripartum antimicrobials	Limited evidence, may reduce risk (Knight, 2019)
Cesarean delivery	Perioperative antimicrobial prophylaxis	Decreased 70–80% (Carter, 2017; Smaill, 2014)
Cesarean delivery	Skin preparation	Decreased incidence (Hadiati, 2018)

The table gives a 70–80% reduction with cesarean prophylaxis, but the text (citing the same Smaill 2014 review) gives 65–75%.

- Benefit applies to **elective and nonelective** cesarean.
- **No proven benefit** after spontaneous vaginal birth, episiotomy repair or manual placental extraction.
- **ACOG: a single dose for 3rd- and 4th-degree lacerations is reasonable** (evidence-based).

Cesarean prophylaxis — the regimen

- **Single 2-g dose of a 1st-generation cephalosporin (cefazolin)** — as effective as broad-spectrum or multidose regimens.
- **Obese women: 3 g cefazolin** for adequate tissue levels.
- **± Azithromycin** → ↓ postcesarean uterine infection.
- **MRSA colonized:** add **vancomycin** to the cephalosporin.

- **Timing:** ACOG favors **before skin incision** (vs after cord clamping; still debated).
- **Penicillin allergy:** most are not at risk of anaphylaxis → **cephalosporin is safe** without an anaphylaxis history. With anaphylaxis → **vancomycin + clindamycin + gentamicin**.

Skin and vaginal preparation

- **Chlorhexidine–alcohol > iodine–alcohol** for surgical-site infection.
- **Vaginal cleansing** with povidone-iodine or **metronidazole gel** may add benefit.

Other prophylaxis — what does and doesn't work

Helps	No effect
Spontaneous placental separation	Treating asymptomatic BV or Trichomonas in pregnancy (no ↓ puerperal infection)
Exteriorizing the uterus for hysterotomy closure	Changing gloves after placental delivery
Subcutaneous closure in obese women → ↓ wound separation (not infection)	Wiping the uterine cavity; dilating the lower segment/cervix
	Single- vs two-layer uterine closure
	Peritoneal closure vs nonclosure

- **Staples vs suture:** staples → more **noninfectious skin separation**.

3. Complications of Uterine and Pelvic Infections

- **>90%** respond within 48–72 h. In the rest: **wound infection, phlegmon or abscess, septic pelvic thrombophlebitis**. All are reduced by prophylaxis.

Abdominal incisional infections

- **A common cause of persistent fever** in women treated for metritis.
- Risk factors: **obesity, diabetes**, corticosteroids, immunosuppression, anemia, hypertension, **hematoma** (poor hemostasis).
- Incidence with prophylaxis **2–10%**; Parkland **~2%**, rising with BMI.
- **Presentation:** persistent fever or fever starting around **day 4**; erythematous wound draining pus.
- Organisms = those in amniotic fluid at cesarean, ± hospital-acquired pathogens.
- **Treatment:**
 - Antibiotics + **surgical drainage and debridement** (usually spinal or general anesthesia); **check fascial integrity**.
 - **Twice-daily wound care**, analgesia tailored to wound (± topical lidocaine), debride and repack with moist gauze.
 - **Granulation by 4–6 days** → **secondary en bloc closure:** polypropylene or nylon suture entering **2–3 cm** from one edge, through the full thickness, exiting **3 cm** from the other edge; remove sutures on **day 10**.

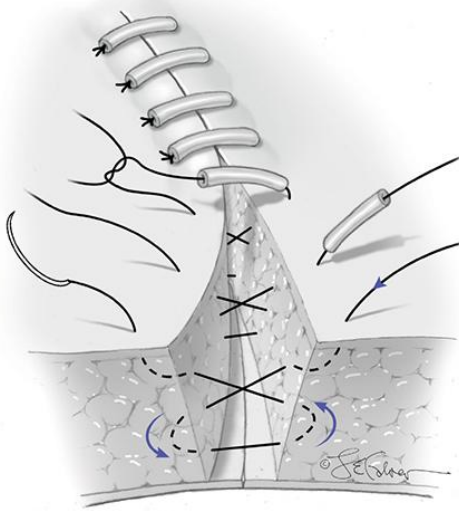


Figure 37-1. Secondary en bloc closure of an open abdominal wound: full-thickness interrupted sutures placed in series.

Negative-pressure wound therapy (NPWT; VAC, topical negative pressure)

- ↑ wound blood flow, angiogenesis, cellular proliferation; ↓ wound size. Widely used despite **meager formal evidence**.
- Uses: open wounds after infection clears, the **"open abdomen"** (most efficient temporary abdominal closure), and perineal wounds (infected episiotomy, hematoma, abscess).
- **No trials** compare it with conventional care after cesarean.

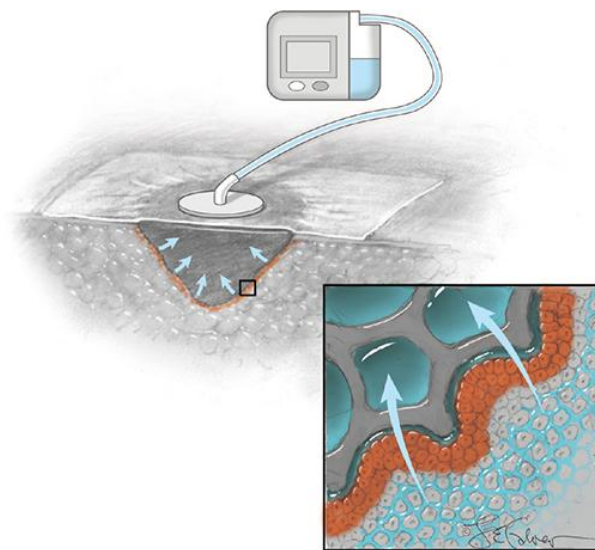


Figure 37-2. NPWT: suction through a porous sponge causes macro- and microdeformation, removes tissue fluid and keeps the wound warm and moist; granulation tissue forms at the sponge interface.

Prophylactic incisional NPWT (obese women)

- **Hussamy (Parkland, 441 morbidly obese women)**: wound morbidity **similar** with iNPWT vs standard dressing (Table 37-4).
- **Denmark (Hyldig, 876 obese women)**: SSI **4.6% vs 9.2%** with NPWT.
- Systematic review: ↓ wound complications. Cost-effectiveness inconclusive.
- **Williams: routine prophylactic NPWT needs more evaluation.**

Table 37-4. Randomized Trial of Prophylactic Incisional Negative-Pressure Wound Therapy (iNPWT) Versus Standard Surgical Dressing for Morbidly Obese Women Undergoing Cesarean Delivery

Factor	iNPWT (N = 222)	Standard (N = 219)	Significance
BMI at delivery	46.6 ± 6.0	45.8 ± 5.8	NS
Cesarean delivery			
Primary	43%	37%	<i>p</i> = 0.18
Secondary	57%	63%	<i>p</i> = 0.18
Scheduled	32%	33%	<i>p</i> = 0.99
Unscheduled	68%	67%	<i>p</i> = 0.99
Pfannenstiel	23%	28%	<i>p</i> = 0.29
Vertical midline	77%	72%	<i>p</i> = 0.29
Incision			
Depth	5.5 ± 1.7	5.3 ± 1.8	<i>p</i> = 0.19
Length	14.5 ± 2.5	14.6 ± 2.9	<i>p</i> = 0.74
Wound morbidity			
Cellulitis	32%	38%	RR = 0.7 (95% CI, 0.3–1.7)
Superficial SSI	54%	61%	RR = 0.8 (95% CI, 0.4–1.5)
Dehiscence	11%	2%	RR = 3.9 (95% CI, 0.4–194)
Composite outcome	17%	19%	RR = 0.9 (95% CI, 0.5–1.4)
Readmission	5%	4%	RR = 1.3 (95% CI, 0.5–3.5)
Reoperation	6%	5%	RR = 1.4 (95% CI, 0.6–3.5)

BMI = body mass index; CI = confidence interval; NS = not significant; RR = relative risk; SSI = surgical site infection. Data from Hussamy, 2018, 2019. The wound-morbidity rows (cellulitis, SSI, dehiscence) appear to be percentages of the women who had wound morbidity, not of the whole group: they are larger than the 17–19% composite rate.

Fascial dehiscence

- Serious; may come with **bowel evisceration**. Risk factors: **wound infection and obesity**.
- **~1 per 300** cesareans (McNeeley, ~9000 women); **2/3 of the 27 cases** had fascial infection and necrosis.
- Usually presents **days 7–10**: superficial separation with **profuse peritoneal fluid or pus leakage**. CT if unclear (only if fast).
- **Dehiscence is a surgical emergency**:
 - Cover or gently replace eviscerated bowel with **saline-soaked sterile towels/gauze**.
 - **Broad-spectrum antibiotics**; exam under anesthesia.
 - Surgery: assess bowel, debride, close fascia with **interrupted mass closure, no. 2 permanent suture**.
 - With infection, leave the subcutaneous layers to close secondarily. Involve general surgery if bowel ischemia or difficult closure is expected.

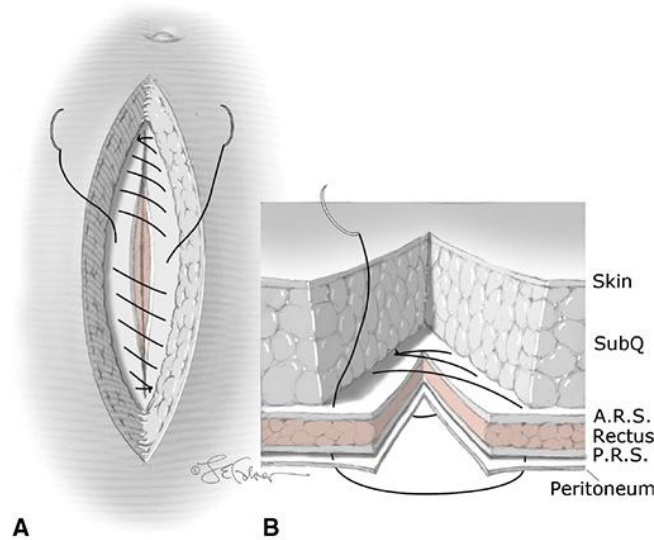


Figure 37-3. Mass closure: each bite 1.5–2 cm from the edge, through peritoneum, rectus muscle and rectus sheath; stitches 1 cm apart.

Necrotizing fasciitis

- Uncommon, **high mortality**; abdominal incisions, episiotomy or perineal lacerations.
- Risk factors increasingly common in gravidas: **diabetes, obesity, hypertension**.
- Usually **polymicrobial** vaginal flora; sometimes a single virulent species (**group A β -hemolytic strep**) or rare pathogens.
- Goepfert: **9 cases in >5000 cesareans**, 2 deaths. Schorge: 5 cases, none with risk factors, none died.
- Can involve skin, superficial and deep subcutaneous tissue, any abdominopelvic fascial layer, $\pm$ muscle (**myofasciitis**).
- **Symptoms usually at 3–5 days**; hard to distinguish from a superficial wound infection $\rightarrow$ septicemia if it progresses.
- **⚠ High suspicion + early surgical exploration if uncertain can be lifesaving.**
- **Treatment**: early diagnosis, **surgical debridement to wide margins of healthy bleeding tissue** (may include abdominal, vulvar, thigh or buttock fascia), antibiotics, ICU care. Mesh may later be needed.
- **Mortality virtually 100% without surgery; ~50% even with exhaustive debridement.**

Peritonitis and adnexal abscesses

- **After cesarean**: infrequent; almost always preceded by **metritis**, especially **uterine incisional necrosis and dehiscence**. Also ruptured adnexal abscess, bowel injury, perforated appendicitis $\rightarrow$ usually **prompt surgery**.
- **After vaginal birth**: rare; often **virulent group A strep**.
- **⚠ Abdominal rigidity may be absent** (lax postpartum abdominal wall). First signs are often those of **adynamic ileus**; marked bowel distension (unusual after vaginal birth).
- Infection spreading from an **intact uterus** $\rightarrow$ antibiotics alone usually suffice.
- **Ovarian abscess**: rare, via a break in the ovarian capsule; usually **unilateral**, presents **1–2 wk** postpartum; **rupture common** $\rightarrow$ severe peritonitis.

Parametrial phlegmon

- Intense parametrial cellulitis after **cesarean** metritis $\rightarrow$ **induration within the broad ligament leaves**.
- **Suspect when fever persists >72 h despite IV antibiotics.**
- Usually **unilateral**, at the base of the broad ligament. Spreads along cleavage planes: most often **laterally to the pelvic sidewall**; sometimes posteriorly into the **rectovaginal septum** (firm mass behind the cervix).
- **Most improve with continued broad-spectrum antibiotics; fever resolves in 5–7 days.**
- Severe cellulitis of the uterine incision $\rightarrow$ **necrosis and separation** $\rightarrow$ pus extrusion, abscess, peritonitis.

- **Surgery only for suspected uterine incisional necrosis:** usually **hysterectomy + debridement**. Difficult (cervix and lower segment inflamed to the sidewall); **adnexa seldom involved** → **ovaries usually conserved**; blood loss often large, transfusion common.

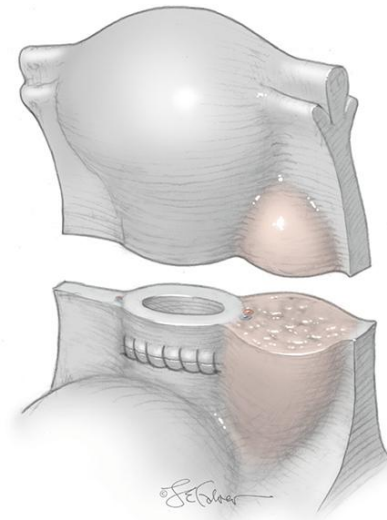


Figure 37-4. Left parametrial phlegmon: indurated cellulitis in the parametrium next to the hysterotomy.

Imaging (CT or MR)

- Brown: CT after **5 days** of refractory infection → abnormal in **75%**, mostly **nonsurgical** lesions.
- Fishel Bartal: refractory fever **>3 days** → abnormal CT in **~60%**; fluid collections **22%**; surgery prompted in **8%**.
- **Imaging mainly helps avoid unnecessary surgery.**
- CT may show incisional dehiscence, but apparent defects (edema) are also seen after uncomplicated cesarean → interpret clinically.

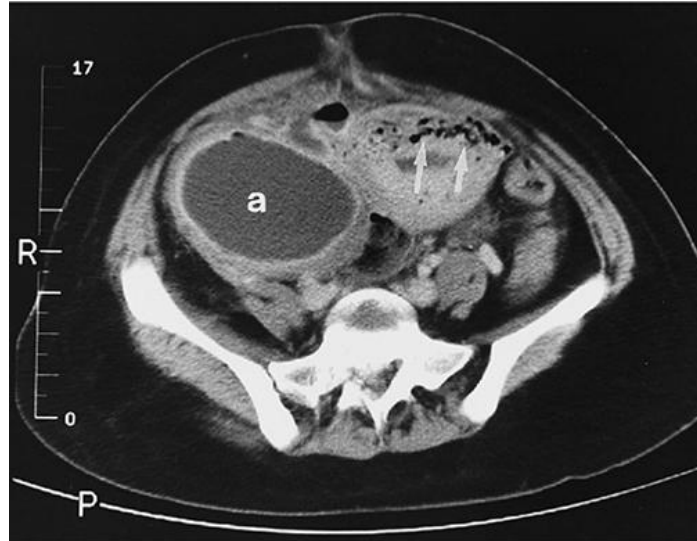


Figure 37-5. CT: necrosis of the uterine incision with gas in the myometrium (arrows) and a large right parametrial abscess (a).



Figure 37-6. Necrotic, dehiscence hysterotomy that leaked into the peritoneal cavity; hysterectomy was needed.

Drainage options

- Suppurated phlegmon → **broad ligament abscess** that may point **above the inguinal ligament** → **CT-guided needle drainage**.
- **Psoas abscess** (rare): percutaneous drainage often needed.
- **Rectovaginal-septum abscess** → **colpotomy drainage**.
- Induration may take days to weeks to resolve.

Septic pelvic thrombophlebitis

- Infection spreads along veins → thrombosis (± lymphangitis). The **ovarian veins** drain the upper uterus/placental site → **one or both ovarian venous plexuses** usually involved.
- Clot extends into the **IVC in ¼**, occasionally the renal vein.

Study	Incidence
Brown (Parkland, 45,000 births)	1/9000 vaginal, 1/800 cesarean
Rouse (16,650 primary cesareans)	1/400; 1/175 with chorioamnionitis, 1/500 without

- **Symptoms are few:** chills, occasional lower-quadrant pain. Women **feel better on antibiotics but stay febrile**.
- **Diagnosis: pelvic CT or MR.**
 - **20%** of 69 women febrile after **>5 days** of antibiotics for metritis (Brown).
 - **6%** of women with refractory fever ≥3 days (Fishel Bartal).
- **Heparin is controversial:**
 - RCT (**14 women**): adding heparin did **not** hasten recovery or improve outcome.
 - Williams: **heparin unnecessary**; others still anticoagulate.
 - **No evidence for long-term anticoagulation.**

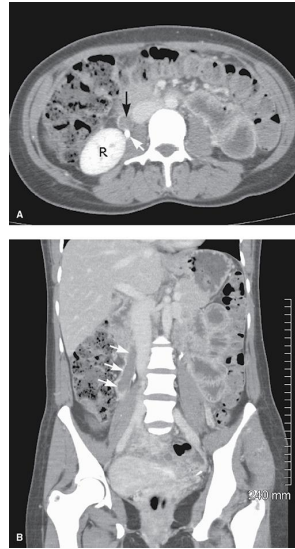


Figure 37-7. Contrast CT: enlarged right ovarian vein filled with low-density thrombus (axial, A; coronal, B). R = right kidney.

4. Perineal Infections

- **Episiotomy infection is uncommon** (fewer episiotomies): **10 in 20,000** vaginal births (Owen).
- **Dehiscence:**
 - Parkland **0.5%**, and **80% of those were infected**. Uygur **1%**, $\frac{2}{3}$ from infection.
- **Anal sphincter tears → more infection: 20%** infection rate (Lewicky-Gaup); influenced by intrapartum antibiotics.
- **Fourth-degree lacerations (Goldaber, 390 women):** morbidity **5.4%** — infection + dehiscence 2.8%, dehiscence only 1.8%, infection only 0.8%.

Pathogenesis and clinical course

- Other dehiscence factors: **coagulation disorders, smoking, HPV infection**. **No data link dehiscence to faulty repair**.
- Symptoms: local pain, **dysuria ± urinary retention**. Commonest findings (Ramin): **pain 65%, purulent discharge 65%, fever 44%**.
- Extreme: whole vulva edematous, ulcerated, covered with exudate. Septic shock or necrotizing fasciitis is rare.
- **Vaginal lacerations:** red, swollen → necrosis and slough; parametrial extension → lymphangitis.
- **Cervical lacerations** seldom look infected; they present as **metritis**. Deep tears into the broad ligament base → lymphangitis, parametritis, bacteremia.

Treatment

- **Cellulitis, no pus:** observation + broad-spectrum antibiotics.
- **Pus:** open, **remove sutures, debride**. **Dehiscence:** local wound care + IV antibiotics.
- **Early repair (after infection clears) — Hauth/Hankins:**
 - Success **94%**; mean **6 days** from dehiscence to repair. Two failures = **pinpoint rectovaginal fistula** (small rectal flap).
- Repair when the wound is **free of exudate with pink granulation tissue**. Mobilize tissue, **identify and mobilize the anal sphincter**, **tension-free** layered closure.
- **Aftercare:** local wound care, **stool softeners, nothing per vagina or rectum** until healed. Aim for **soft formed stools** (hard stool disrupts; liquid stool seeps in and reinfects).

Table 37-5. Preoperative Protocol for Early Repair of Episiotomy Dehiscence

Step	Action
1	Open wound, remove sutures, begin intravenous antimicrobials
2	Initiate wound care: • Institute sitz bath several times daily or hydrotherapy • Provide adequate analgesia or anesthesia — regional analgesia or general anesthesia may be necessary for initial debridements • Scrub wound twice daily with a povidone-iodine solution • Debride necrotic tissue
3	Close wound when afebrile and pink, healthy granulation tissue present
4	Provide enemas prior to fourth-degree repair
5	Institute postoperative stool softeners; normal diet, nothing per vagina or rectum

5. Toxic Shock Syndrome

- Acute febrile multisystem illness; **case-fatality 10–15%**.
- **Features:** fever, headache, confusion, **diffuse macular erythematous rash**, nausea, vomiting, **watery diarrhea**, **marked hemoconcentration** → renal then hepatic failure, **DIC**, circulatory collapse. **Rash areas desquamate** during recovery.

Cause	Mechanism / notes
<i>Staphylococcus aureus</i> (incl. MRSA)	TSST-1 exotoxin → profound endothelial injury; tiny amounts activate T cells → "cytokine storm"
Group A β-hemolytic strep	Pyrogenic exotoxin → streptococcal TSS; serotypes M1 and M3 especially virulent
<i>Clostridium sordellii</i> and <i>C. novyi</i>	Almost identical picture in pregnant women

- Infection may not be apparent — **mucosal colonization** is the presumed source.
 - **10–20%** of pregnant women carry vaginal *S. aureus*; *C. perfringens/sordellii* in **3–10%** of asymptomatic women.
- **Delayed diagnosis** → **maternal death**. Antenatal TSS can cause neonatal death.
- **Treatment:** mainly **supportive** while endothelial injury reverses; antibiotics covering **staphylococci and streptococci** (+ polymicrobial coverage if pelvic infection); **wound debridement ± hysterectomy**. Mortality is high because the toxin is so potent.

6. Breast Infections

Mastitis

- Rare antepartum; **postpartum incidence ~3%**. **No proven prophylaxis**.
- Risk factors: **nursing difficulties**, **cracked nipples**, **oral antibiotic therapy**.
- **Onset rarely before the end of week 1; usually weeks 3–4**.
- Almost always **unilateral**; marked **engorgement precedes** inflammation.
- **Chills or rigors** → **fever and tachycardia**; severe pain; hard, red breast.
- **~10% develop an abscess** — fluctuation is hard to feel → **sonography is usually diagnostic**.
- Rarely, **TSS** from *S. aureus* mastitis.

Etiology

- ***S. aureus*, especially MRSA — the most common isolate** (40% in Matheson). Others: **coagulase-negative staphylococci, viridans streptococci**.
- **Source: the newborn's nose and throat**; bacteria enter through **nipple fissures/abrasions**. Organism usually cultured from milk.
- Epidemics coincide with new resistant staphylococcal strains (now MRSA).
 - Parkland 2000–2004: **¼ of community-acquired MRSA** isolates were from pregnant/postpartum women with mastitis.
 - Hospital MRSA: newborn colonized by nursery staff.
 - **MRSA mastitis → more recurrent abscess**.

Management

- Started before suppuration → **usually resolves within 48 h**.
- Many **culture expressed milk** before treatment (surveillance, sensitivities).
- Best regimen unclear; choice reflects local staphylococcal patterns.

Situation	Antibiotic
Empirical	Dicloxacillin 500 mg orally four times daily
Penicillin sensitivity	Erythromycin
Resistant/penicillinase-producing staph suspected (e.g., MRSA)	Vancomycin, clindamycin or trimethoprim-sulfamethoxazole
Duration	10 days , even if response is prompt

- **⚠ Continue breastfeeding**. Marshall (65 women): the **only 3 abscesses** occurred among the **15 who stopped** breastfeeding. Vigorous milk expression alone may suffice.
- Infant refusing the inflamed breast is due to **engorgement/edema** (harder areolar grip), not milk taste → **pumping** helps.
- **Start suckling on the uninvolved breast** so let-down begins, then move to the tender side.
- **HIV (resource-poor settings)**: breastfeeding is allowed, but **stop feeding from a breast with mastitis or abscess** — HIV RNA rises in that milk (returns to baseline after resolution).

Breast abscess

- Incidence **0.1%** (1.5 million Swedish women).
- **Suspect if no defervescence within 48–72 h** of treatment or a **palpable mass** → **sonography**. Can be large (**2 L** of pus in one report).

Approach	Details
Sonographically guided needle aspiration (now favored; local analgesia)	Success 80–90% . RCT at 8 wk: healed 93% vs 77% with surgical drainage
Surgical drainage (traditional; usually general anesthesia)	Incision along Langer lines ; single incision over the most dependent fluctuant area; multiple abscesses → several incisions + break loculations; loose gauze pack, replaced at 24 h with a smaller one

- Sonographic findings, first antibiotic and organism do **not** predict aspiration failure.
- **Nonhealing abscess** → consider **granulomatous mastitis, cancer, tuberculosis**.

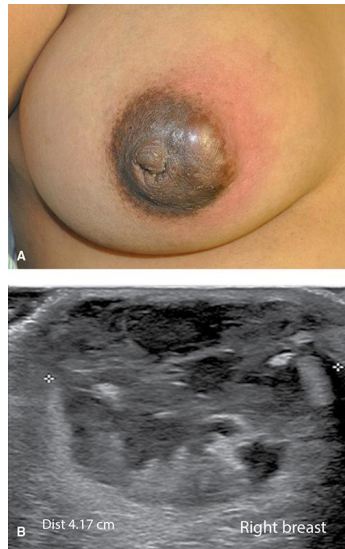


Figure 37-8. Puerperal mastitis with abscess: (A) indurated, erythematous right breast; (B) sonogram of a 5-cm abscess.

Quick-Recall: Persistent Postpartum Fever — Differential

Cause	Timing / clue	Diagnosis	Management
Metritis	Fever 38–39°C, uterine/parametrial tenderness, esp. post-cesarean	Clinical	Clindamycin + gentamicin (± ampicillin); stop after 24 h afebrile
Group A strep	Fever ≥39°C in first 24 h; scant odorless lochia	Blood culture	Urgent broad therapy; high mortality
Wound infection / abscess	Fever ~day 4, red draining wound	Exam	Open, drain, debride; secondary closure at 4–6 d
Fascial dehiscence	Days 7–10, gush of serous/purulent fluid	Exam ± CT	Emergency: mass closure with no. 2 permanent suture
Necrotizing fasciitis	Days 3–5, looks worse than wound	Surgical exploration	Wide debridement (50% mortality even then)
Parametrial phlegmon	Fever >72 h on IV antibiotics; broad-ligament induration	CT/MR	Continue antibiotics (5–7 d); surgery only for incisional necrosis
Pelvic/broad-ligament abscess	Fluctuant mass, may point above inguinal ligament	CT	CT-guided drainage; colpotomy if rectovaginal septum
Septic pelvic thrombophlebitis	Feels well but febrile despite antibiotics	CT/MR: ovarian vein thrombus	Continue antibiotics; heparin optional
Breast engorgement	Non-breastfeeding, <39°C, <24 h	Clinical	Supportive
Mastitis	Week 3–4, unilateral red hard breast	Clinical	Dicloxacillin 10 d; keep breastfeeding
Breast abscess	No response in 48–72 h, mass	Sonography	Needle aspiration (80–90%)
Atelectasis	Post-general anesthesia	Clinical	Coughing, deep breathing
Toxic shock	Rash, diarrhea, hemoconcentration, shock	Clinical	Supportive; anti-staph/strep; debridement

Key Numbers to Memorize

- Fever **≥38.0°C**; infection causes **13.6%** of maternal deaths (2nd); **40%** preventable
- Fever <24 h → pelvic infection **20%** vaginal vs **70%** cesarean; engorgement fever in **15%** non-breastfeeders
- Metritis: vaginal **1–2%** (5–6% with risks, >13% with chorioamnionitis); cesarean **50%** without prophylaxis

- Prophylaxis ↓ infection **65–75%** (text) / **70–80%** (table); **cefazolin 2 g** (3 g if obese); azithromycin extra ↓ **12 → 6%**
- Amnionic fluid: **2.5** organisms; mixed **63%**, anaerobes alone **30%**, aerobes alone **7%**
- Group A strep **1/1220** births; maternal mortality **~90%** in early-onset cases
- Response in **48–72 h** in **~90%**; clindamycin + gentamicin **90–97%**; discharge after **24 h** afebrile
- Wound infection **2–10%** (~2% Parkland); secondary closure at **4–6 d**, sutures out at **10 d**
- Fascial dehiscence **1/300**, days **7–10**; necrotizing fasciitis mortality **~50%** with surgery
- Phlegmon: fever **>72 h**, resolves in **5–7 d**; CT abnormal **75%**, surgery in **8%**
- Septic pelvic thrombophlebitis: **1/9000 vaginal**, **1/800 cesarean** (1/175 with chorioamnionitis); IVC in **1/4**
- Episiotomy dehiscence **0.5–1%**; sphincter-tear infection **20%**; early repair success **94%** at **~6 d**
- TSS fatality **10–15%**; vaginal *S. aureus* in **10–20%**
- Mastitis **3%**, weeks **3–4**; abscess in **10%** of mastitis (**0.1%** overall); dicloxacillin **500 mg qid × 10 d**; aspiration success **80–90%**